Latihan Topik 3 Relational Model (Aljabar Relasional)

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Skema yang dipakai:
loan (loan_number, branch_name, amount)
borrower (customer_name, loan_number)
account (account_number, branch_name, balance)
depositor (customer_name, account_number)

1. Find all loans of over $1200

$$\sigma_{amount > 1200} (loan)$$

2. Find the loan number for each loan of an amount greater than $1200

$$\Pi_{loan_number} (\sigma_{amount > 1200} (loan))$$

3. Find all loans greater than $1200 or less than $75

$$\sigma_{amount > 1200 \vee amount < 75} (loan)$$

4. Find the names of all customers who have a loan, an account, or both, from the bank

$$\Pi_{customer_name} (borrower) \cup \Pi_{customer_name} (depositor)$$

5. Find the names of all customers who have a loan and an account at the bank

$$\Pi_{customer_name} (borrower) \cap \Pi_{customer_name} (depositor)$$

6. Find the names of all customers who have a loan at the Riverside branch

$$\Pi_{customer_name} (\sigma_{branch_name = 'Riverside'} (borrower \bowtie loan))$$

7. Find the names of all customers who have a loan at the Riverside branch but do not have an account at any branch of the bank

$$\Pi_{customer_name} (\sigma_{branch_name = 'Riverside'} (borrower \bowtie loan)) - \Pi_{customer_name} (depositor)$$

8. Find the largest account balance

$$\Im_{MAX\ balance} (account)$$